

Artificial Bee Colony Search for Improving Determinantal Sampling Designs

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Abstract

Determinantal sampling designs provide a flexible class of fixed-size sampling designs with prescribed first-order inclusion probabilities and computable second-order inclusion probabilities. In a fixed-size determinantal design, the sampling kernel is a projection matrix, and the diagonal of the kernel gives the prescribed inclusion probabilities. However, the kernel is not unique: many projection matrices may have the same diagonal but lead to different second-order inclusion probabilities and hence different Horvitz–Thompson variances.

This note proposes an Artificial Bee Colony (ABC) search strategy for improving determinantal sampling designs within a parametrized family of kernels. Each candidate solution is represented by tuning matrices (ω, ρ) , which are mapped to a determinantal kernel $K(\omega, \rho)$. Candidate quality is evaluated by the exact design-based Horvitz–Thompson variance, without Monte Carlo simulation inside the optimizer. The approach is compared with a random-search baseline using the same parametrization. A small exploratory illustration on a 30-unit subset of the MU284 population is included to show the current behavior of the method and to motivate further discussion.

1 Introduction

Unequal-probability sampling designs are widely used when auxiliary information is available and units differ in size, importance, or expected contribution to the population total. Let $\mathcal{U} = \{1, \dots, N\}$ be a finite population and let π_k be the first-order inclusion probability of unit k . For a fixed sample size n , one typically requires

$$0 < \pi_k < 1, \quad \sum_{k=1}^N \pi_k = n.$$

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When all first-order inclusion probabilities are positive, the Horvitz–Thompson estimator (Horvitz and Thompson, 1952)

$$\hat{t}_{y,\text{HT}} = \sum_{k \in S} \frac{y_k}{\pi_k}$$

is unbiased for the population total $t_y = \sum_{k \in \mathcal{U}} y_k$. Its variance, however, depends on the second-order inclusion probabilities. Thus, even after fixing the vector π , different sampling designs may have substantially different efficiencies.

Determinantal sampling designs, introduced in finite-population sampling by Loonis and Mary (2019), are attractive in this setting because their inclusion probabilities are directly available from a kernel matrix. A determinantal design $\text{DSD}(K)$ is defined by

$$\Pr(s \subseteq S) = \det(K_s), \quad s \subseteq \mathcal{U},$$

where K_s is the submatrix of the Hermitian kernel K indexed by s . The first- and second-order inclusion probabilities are

$$\pi_k = K_{kk}, \quad \pi_{kl} = K_{kk}K_{ll} - |K_{kl}|^2, \quad k \neq l.$$

Therefore, the off-diagonal entries of K control pairwise repulsion between sampled units. For a fixed-size determinantal design, K is a projection matrix:

$$K^2 = K, \quad K = K^*, \quad \text{tr}(K) = n.$$

The construction of fixed-size determinantal designs with a prescribed diagonal is closely related to constructive forms of the Schur–Horn problem and to the construction of self-adjoint matrices with prescribed spectrum and diagonal (Fickus et al., 2013). The design proposed by Loonis and Mary (2019), denoted here by P^π , provides a natural reference kernel with $\text{diag}(P^\pi) = \pi$. More general parametrizations of determinantal sampling designs were studied by Loonis (2023).

The present work explores whether this reference design can be improved by a derivative-free optimization procedure. We use the Artificial Bee Colony algorithm, originally introduced by Karaboga (2005) and further developed by Basturk and Karaboga (2006); Karaboga and Basturk (2007); Karaboga and Akay (2009). The ABC algorithm is a population-based metaheuristic inspired by the foraging behavior of honey bees. In our setting, each food source represents a candidate pair of matrices (ω, ρ) , which is transformed into a determinantal kernel. The best kernel is selected according to the efficiency of the corresponding Horvitz–Thompson estimator.

This note is intended as a cleaned exploratory version for discussion. The numerical experiment is deliberately small and is not presented as a final journal-scale simulation study.

2 Determinantal sampling framework

Let $y = (y_1, \dots, y_N)^\top$ be the study variable and let $z = (z_1, \dots, z_N)^\top$ be a target or auxiliary variable used for optimization. For a determinantal kernel K , define

$$A(K) = K \circ (I_N - \overline{K}),$$

where $\circ$ denotes the Hadamard product. The Horvitz–Thompson variance under $\text{DSD}(K)$ can be computed as

$$\text{Var}_K(\widehat{t}_{y,\text{HT}}) = y^* D_\pi^{-1} A(K) D_\pi^{-1} y, \quad D_\pi = \text{diag}(\pi_1, \dots, \pi_N).$$

In the real-valued case this reduces to the usual quadratic form with transpose instead of conjugate transpose.

We compare the candidate design with simple random sampling without replacement of the same size n . The SRS variance for the total is

$$\text{Var}_{\text{SRS}}(\widehat{t}_y) = N^2 \left(1 - \frac{n}{N}\right) \frac{s_y^2}{n},$$

where s_y^2 is the finite-population variance computed with denominator $N - 1$. The efficiency of a kernel K for a variable u is

$$\text{Eff}_u(K) = \frac{\text{Var}_{\text{SRS}}(\widehat{t}_u)}{\text{Var}_K(\widehat{t}_{u,\text{HT}})}.$$

Larger values indicate lower variance relative to SRS.

The reference determinantal design is denoted by P^π . Its kernel is constructed so that

$$\text{diag}(P^\pi) = \pi, \quad P^{\pi^2} = P^\pi, \quad \text{tr}(P^\pi) = n.$$

For comparison with P^π , we use the ratio

$$R_u(K) = \frac{\text{Eff}_u(K)}{\text{Eff}_u(P^\pi)}, \quad u \in \{y, z\}.$$

Thus $R_u(K) > 1$ means that K is more efficient than the reference design P^π for variable u , while $R_u(K) < 1$ means that it is less efficient.

3 ABC search over parametrized determinantal kernels

The search is performed over a parametrized family of kernels rather than over the entries of K directly. For a given inclusion-probability vector π , the implementation uses a mapping

$$\Psi_\pi : (\omega, \rho) \mapsto K(\omega, \rho),$$

where

$$\omega \in [0, 1]^{M \times N}, \quad \rho \in [0, 1]^{M \times (N-1)}.$$

The function `CaDsd` constructs the candidate kernel from (π, M, ω, ρ) . The matrix ω controls spectral choices, while ρ controls rotation or phase-like transformations used in the kernel construction. Each candidate is evaluated through the exact variance formula above.

In the ABC implementation, a food source is a pair $F_i = (\omega_i, \rho_i)$. Its score is

$$Q(F_i) = \text{Eff}_z\{K(\omega_i, \rho_i)\},$$

although other objectives, such as Eff_y or a combined mean/harmonic criterion, can also be used. The employed-bee and onlooker-bee phases generate perturbations of existing food sources, while scout moves replace sources that fail to improve. A small local-search step around the best current source is also used. Elitism ensures that the best source found so far is retained.

A random-search baseline is run in parallel. It uses the same **CaDsd** parametrization, the same validity checks, and the same variance calculation, but it generates candidates independently rather than adaptively. Therefore, the comparison between ABC and random search separates the effect of the colony mechanism from the effect of merely sampling many random kernels.

The reference ordering follows the Loonis–Mary construction, where units are ranked by z_k/π_k before constructing the reference P^π design. The ABC class then internally reorders units by decreasing π_k when evaluating **CaDsd**, because the **CaDsd** implementation works in that internal order. All variance calculations are performed after aligning y , z , and π consistently with the kernel order.

4 Exploratory numerical illustration

We used a 30-unit subset of the MU284 population. This small run is intended only as a preliminary comparison for discussion. The study variable is

$$y = \text{P85}.$$

The optimization variables are

$$z \in \{\text{ME84}, \text{S82}, \text{CS82}\}.$$

Three inclusion-probability settings are considered: unequal probabilities proportional to P75, unequal probabilities proportional to SS82, and equal probabilities. The sample size is fixed at $n = 5$.

For each case, the ABC algorithm was run for 1000 iterations with colony size 20, abandonment limit 25, onlooker factor 0.55, local search every 12 iterations, and objective function Eff_z . The random-search baseline used 20 initial evaluations and 31 evaluations per ABC iteration. The table reports ratios relative to the Vincent ordered P^π reference. The notation is as follows:

$$R_z = \frac{\text{Eff}_z(K)}{\text{Eff}_z(P^\pi)}, \quad R_y = \frac{\text{Eff}_y(K)}{\text{Eff}_y(P^\pi)}.$$

The columns $R_{z,0}$ and $R_{y,0}$ give common starting ratios, computed as the average of the ABC and random-search ratios at iteration 1. The columns $R_{z,\text{ABC}}$, $R_{y,\text{ABC}}$, $R_{z,\text{RND}}$, and $R_{y,\text{RND}}$ give the final ratios at iteration 1000. The symbols $\rho(y, \pi)$ and $\rho(y, z)$ denote the empirical correlations between y and the inclusion-probability size variable, and between y and the optimization variable z , respectively. In the equal-probability case, $\rho(y, \pi)$ is undefined because all inclusion probabilities are constant.

The exploratory results show a clear difference between the adaptive ABC search and the random-search baseline. In all nine cases, the final z -efficiency ratio of ABC is larger than

Table 1: Exploratory comparison of ABC and random search relative to the Vincent ordered P^π reference. The population size is $N = 30$ and the sample size is $n = 5$. The columns $R_{z,0}$ and $R_{y,0}$ report common starting ratios, computed as the average of the ABC and random-search ratios at iteration 1. The final columns report the ratios at iteration 1000. Values larger than 1 indicate improvement over the Vincent P^π reference.

z	π	$\rho(y, \pi)$	$\rho(y, z)$	$R_{z,0}$	$R_{z,ABC}$	$R_{z,RND}$	$R_{y,0}$	$R_{y,ABC}$	$R_{y,RND}$
ME84	P75	0.98	0.98	0.17	0.56	0.19	0.87	1.27	0.85
S82	P75	0.98	0.95	0.36	1.05	0.43	1.06	1.45	1.03
CS82	P75	0.98	0.79	0.32	0.94	0.40	0.86	1.03	0.85
ME84	SS82	0.63	0.98	0.14	0.61	0.16	0.13	0.47	0.14
S82	SS82	0.63	0.95	0.19	0.61	0.23	0.74	1.12	0.81
CS82	SS82	0.63	0.79	0.18	0.49	0.21	0.40	1.35	0.36
ME84	Equal	–	0.98	0.46	1.00	0.69	0.39	1.00	0.63
S82	Equal	–	0.95	0.47	1.12	0.66	0.40	0.89	0.59
CS82	Equal	–	0.79	0.14	1.02	0.17	0.55	1.00	0.55

the corresponding random-search ratio. For example, with equal inclusion probabilities and $z = \text{S82}$, ABC moves from an initial ratio of about 0.47 to 1.12, while random search ends at 0.66. With $z = \text{CS82}$ and equal inclusion probabilities, ABC moves from 0.14 to 1.02, while random search remains at 0.17. These cases show that the colony mechanism is not merely reproducing random exploration; it is able to exploit information gathered during the search.

The unequal-probability cases are more mixed. When π is proportional to P75, the reference design is already very strong for variables highly correlated with y and with the size variable. This explains why ABC does not exceed the reference for $z = \text{ME84}$, although it still improves substantially over the starting point and over random search. For $z = \text{S82}$, ABC reaches a ratio above one for both z and y . When π is proportional to SS82, the z -ratios remain below one, but ABC still gives a large improvement over random search and, for $z = \text{S82}$ and $z = \text{CS82}$, it improves the y -efficiency relative to P^π .

Numerical validity of the optimized kernels. For each optimized kernel, we checked the main numerical properties required by a fixed-size determinantal sampling design: trace, numerical rank, diagonal entries, projection error, eigenvalue range, and second-order inclusion probabilities. In the reported runs, the optimized kernels preserved the prescribed first-order inclusion probabilities with errors of order 10^{-3} , had numerical rank equal to the target sample size $n = 5$, and produced nonnegative second-order inclusion probabilities.

The kernels did not satisfy the strict projection conditions at machine precision. For example, the trace was close to, but not exactly equal to, 5, and the largest eigenvalue was slightly larger than 1 in some cases. This occurred because the search was run in a fast validation mode. In that mode, candidates were mainly screened by checking whether the diagonal matched the target inclusion probabilities, while the exact projection condition $K^2 = K$, the exact trace condition $\text{tr}(K) = n$, and the exact eigenvalue condition were not enforced at every candidate evaluation. Therefore, these results should be interpreted as exploratory numerical evidence based on kernels that are close to valid fixed-size determinantal kernels, but not exact projection kernels. For a journal version, the optimization should be run with

stricter projection-based validation, or followed by a projection/repair step to ensure that the final kernel lies exactly on the fixed-size DSD space.

5 Discussion and next steps

The preliminary results suggest that ABC can improve determinantal kernels within the current parametrization and that its improvement is not simply due to random candidate generation. The comparison with random search is especially useful because both methods use the same kernel constructor, the same inclusion probabilities, and the same variance objective. The observed gains therefore support the value of the adaptive search mechanism.

At the same time, the current experiment is intentionally limited. The population size is only $N = 30$, the sample size is fixed at $n = 5$, and only a small number of variable and inclusion-probability settings are considered. The final optimized kernels were also obtained under a fast numerical validation mode. These features make the results suitable for methodological discussion but not yet sufficient for a final simulation section.

The next stage should focus on three improvements. First, the kernel validation step should be made stricter so that the final accepted designs are exact, or near machine-precision, fixed-size projection kernels. Second, the experiment should be repeated on larger populations and additional sample sizes. Third, the comparison should be extended to established competitors, such as balanced sampling by the cube method (Deville and Tillé, 2004), in order to compare the determinantal, repulsion-based approach with balance-based sampling.

6 Conclusion

This note presents an ABC-based search strategy for improving determinantal sampling designs with prescribed first-order inclusion probabilities. The method searches over a parametrized family of kernels and uses exact Horvitz–Thompson variance as the objective. In a small exploratory MU284 experiment, ABC consistently outperformed a random-search baseline in terms of final z -efficiency ratios relative to the Vincent ordered P^π reference. The results are promising but preliminary. A final version should enforce stricter projection validity and expand the simulation study before journal submission.

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